1. PRODUCT AND COMPANY IDENTIFICATION

Trade name
NIPACIDE CI 15 HS 0025

Material number: 227943

Substance name: Biocidal preparation

Recommended use:
Industry sector : Polymers industry
Type of use: Biocidal product

Supplier name/address/telephone no.:
Clariant Specialty Chemicals
(Zhenjiang) Co. Ltd.
39,Linjiang Xi Road,New District,Zhenjiang City,Jiangsu Pro.
Zhenjiang City 212132, China
Telephone : +86 511 8595 6637

Information about the substance/preparation
Product Safety : BU Care Chemicals
Product Stewardship
e-mail: SDS.China@clariant.com

Emergency telephone number: +65 3158 1074

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Inhalation)	:	Category 4
Skin corrosion/irritation	:	Category 1
Serious eye damage/eye irritation	:	Category 1
Skin sensitisation	:	Category 1
Long-term (chronic) aquatic hazard	:	Category 3

GHS label elements

Hazard pictograms :

Signal word : Danger

Hazard statements : H314 Causes severe skin burns and eye damage.
H317 May cause an allergic skin reaction.

H332 Harmful if inhaled.
H412 Harmful to aquatic life with long lasting effects.

Precautionary statements :

Prevention:

P261 Avoid breathing mist or vapours.
P264 Wash skin thoroughly after handling.
P271 Use only outdoors or in a well-ventilated area.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower.
P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P362 + P364 Take off contaminated clothing and wash it before reuse.

Storage:

P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
Sodium chlorate	7775-09-9	< 10
5-Chloro-2-methyl-2H-isothiazol-3-one	26172-55-4	>= 1 -< 2,5
2-Methylisothiazolin-3-one	2682-20-4	>= 0,25 -< 1
Copper sulphate	7758-98-7	>= 0,025 -< 0,25

4. FIRST AID MEASURES

General advice	:	Move out of dangerous area. Consult a physician. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended.
If inhaled	:	Move to fresh air. If unconscious, place in recovery position and seek medical advice. Never give anything by mouth to an unconscious person. Get immediate medical advice/ attention.
In case of skin contact	:	If on skin, rinse well with water. Take off all contaminated clothing immediately. Take victim immediately to hospital.
In case of eye contact	:	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get immediate medical advice/ attention. If easy to do, remove contact lens, if worn. Protect unharmed eye. Keep eye wide open while rinsing.
If swallowed	:	Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Take victim immediately to hospital.
Most important symptoms and effects, both acute and delayed	:	The possible health hazards known are those derived from the labelling (see corresponding section) and/or provided in this section. The possible symptoms known are those derived from the labelling (see section 2).
Notes to physician	:	Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	Water spray jet Alcohol-resistant foam Dry powder Carbon dioxide (CO_2) Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	:	High volume water jet
Hazardous combustion products	:	Chlorine compounds Carbon oxides Nitrogen oxides (NOx)

		Sulphur compounds Hydrocarbons Metal oxides Sodium oxides
Specific extinguishing methods	:	In the event of fire and/or explosion do not breathe fumes. Do not allow run-off from fire fighting to enter drains or water courses. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Special protective equipment for firefighters	:	Wear full protective clothing and self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Avoid contact with skin, eyes and clothing. Ensure adequate ventilation. Evacuate personnel to safe areas. Remove all sources of ignition. Use personal protective equipment. Refer to protective measures listed in sections 7 and 8.
Environmental precautions	:	The product should not be allowed to enter drains, water courses or the soil.
Methods and materials for containment and cleaning up	:	Prevent product from entering drains. Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Clean contaminated surface thoroughly. Treat recovered material as described in the section "Disposal considerations".

7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Keep away from heat and sources of ignition. Observe the general rules of industrial fire protection
Advice on safe handling	:	Use only with adequate ventilation/personal protection. For personal protection see section 8. Avoid contact with skin, eyes and clothing. Do not breathe dust/ fume/ gas/ mist/ vapours/ spray. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
Conditions for safe storage	:	Keep containers tightly closed in a dry, cool and well-ventilated place. Handle and open container with care. Keep away from sources of ignition - No smoking.

Further information on storage conditions	:	Store at a temperature between 5 to 40 °C
Materials to avoid	:	When used and handled as intended, none.
Storage period	:	12 Months
Further information on storage stability	:	Stable under recommended storage conditions.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures	:	Use engineering controls such as local or general exhaust to maintain airborne concentrations below exposure limits.

Personal protective equipment

Respiratory protection	:	Use respiratory protection unless adequate local exhaust ventilation is provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines.
Filter type	:	Combined particulates, inorganic and acidic gas/vapour, ammonia/amines and organic vapour type
Hand protection		
Remarks	:	Chemical resistant gloves Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time.
Break through time	:	30 min
Glove thickness	:	0,4 mm For short-term exposure (splash protection): Nitrile rubber gloves.
Break through time	:	480 min
Glove thickness	:	0,7 mm Long-term exposure Impervious butyl rubber gloves
Eye protection	:	Chemical splash goggles with face shield. Do not wear contact lenses.
Skin and body protection	:	Dermal contact should be prevented through the use of impervious clothing, footwear, and a face shield where splattering may occur.
Protective measures	:	Observe the usual precautions for handling chemicals.

		Avoid breathing dust or vapour.
Hygiene measures	:	Handle in accordance with good industrial hygiene and safety practice. Use protective skin cream before handling the product. Wash hands before breaks and at the end of workday. Take off immediately all contaminated clothing and wash it before reuse.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	:	Liquid
Colour	:	see product information sheet
Odour	:	odourless
Odour Threshold	:	no data available
pH	:	2 - 4 (20 °C) Concentration: 100 %
Melting point	:	approx. 0 °C
Boiling point	:	approx. 100 °C
Flash point	:	Not applicable
Evaporation rate	:	no data available
Self-ignition	:	no data available
Upper explosion limit / upper flammability limit	:	no data available
Lower explosion limit / Lower flammability limit	:	no data available
Vapour pressure	:	no data available
Relative vapour density	:	no data available
Relative density	:	no data available
Density	:	approx. 1,06 g/cm3 (20 °C)
Solubility(ies)		
Water solubility	:	soluble (20 °C)

Partition coefficient: n-octanol/water	:	Not applicable
Auto-ignition temperature	:	no data available
Decomposition temperature	:	200 °C Heating rate: 3 K/min Method: DSC No decomposition if used as directed.
Viscosity		
Viscosity, dynamic	:	20 mPa.s (20 °C) Method: Brookfield
Viscosity, kinematic	:	no data available
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.
Metal corrosion rate	:	no data available
Particle size	:	Does not apply to liquids.

10. STABILITY AND REACTIVITY

Reactivity	:	No dangerous reaction known under conditions of normal use.
Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	Reactions with oxidising agents. Reactions with reducing agents. Reactions with amines.
Conditions to avoid	:	None known.
Incompatible materials	:	not known
Hazardous decomposition products	:	When handled and stored appropriately, no dangerous decomposition products are known

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Harmful if inhaled.

Product:

Acute oral toxicity	:	LD50 (Rat): > 2.000 mg/kg Method: Calculation method
Acute inhalation toxicity	:	Acute toxicity estimate: 3,39 mg/l Exposure time: 4 h Test atmosphere: dust/mist

		Method: Calculation method
Acute dermal toxicity	:	Acute toxicity estimate: > 2.000 mg/kg Method: Calculation method

<u>Components:</u>

Sodium chlorate:

Acute oral toxicity	:	Assessment: The component/mixture is toxic after single ingestion.

5-Chloro-2-methyl-2H-isothiazol-3-one:

Acute oral toxicity	:	Assessment: The component/mixture is toxic after single ingestion.
Acute inhalation toxicity	:	Assessment: The component/mixture is highly toxic after short term inhalation.
Acute dermal toxicity	:	Assessment: The component/mixture is toxic after single contact with skin.

2-Methylisothiazolin-3-one:

Acute oral toxicity	:	LD50 (Rat): 285,5 mg/kg Method: OECD Test Guideline 401
Acute inhalation toxicity	:	LC50 (Rat): 0,11 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Assessment: Corrosive to the respiratory tract.
Acute dermal toxicity	:	LD50 (Rat): > 2.000 mg/kg Assessment: The component/mixture is toxic after single contact with skin.

Copper sulphate:

Acute oral toxicity	:	Assessment: The component/mixture is moderately toxic after single ingestion.

Skin corrosion/irritation

Causes severe burns.

<u>Components:</u>

5-Chloro-2-methyl-2H-isothiazol-3-one:

Result	:	Causes burns.

2-Methylisothiazolin-3-one:

Species	:	Rabbit

Method	:	OECD Test Guideline 404
Result	:	Causes burns.

Copper sulphate:

Result	:	Irritating to skin.

Serious eye damage/eye irritation

Causes serious eye damage.

<u>Product:</u>

Species	:	Rabbit
Result	:	Corrosive

<u>Components:</u>

2-Methylisothiazolin-3-one:

Result	:	Risk of serious damage to eyes.
Remarks	:	By analogy with a product of similar composition

Copper sulphate:

Result	:	Irritating to eyes.

Respiratory or skin sensitisation

Skin sensitisation

May cause an allergic skin reaction.

Respiratory sensitisation

Not classified

<u>Product:</u>

Species	:	Guinea pig
Result	:	Causes sensitisation.

<u>Components:</u>

5-Chloro-2-methyl-2H-isothiazol-3-one:

Result	:	May cause sensitisation by skin contact.

2-Methylisothiazolin-3-one:

Test Type	:	Buehler Test
Exposure routes	:	Dermal
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	The product is a skin sensitiser, sub-category 1A.
Assessment	:	Toxic if swallowed., Fatal if inhaled., Causes severe skin burns and eye damage. May cause an allergic skin reaction.

Germ cell mutagenicity

Not classified

Components:

2-Methylisothiazolin-3-one:

Genotoxicity in vitro	:	Test Type: Ames test Metabolic activation: with and without metabolic activation Result: negative Test Type: Chromosome aberration test in vitro Test system: mammalian cells Metabolic activation: with and without metabolic activation Result: negative Test Type: Micronucleus test Test system: mammalian cells Metabolic activation: with and without metabolic activation Result: negative
Germ cell mutagenicity - Assessment	:	In vitro tests did not show mutagenic effects, In vivo tests did not show mutagenic effects

Carcinogenicity

Not classified

Components:

2-Methylisothiazolin-3-one:

Carcinogenicity - Assessment	:	Not classifiable as a human carcinogen.

Reproductive toxicity

Not classified

Components:

2-Methylisothiazolin-3-one:

Effects on fertility	:	Remarks: This information is not available.
Effects on foetal development	:	Remarks: Based on available data, the classification criteria are not met.
Reproductive toxicity - Assessment	:	No evidence of adverse effects on sexual function and fertility, or on development, based on animal experiments.

STOT - single exposure

Not classified

Components:

2-Methylisothiazolin-3-one:

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure

Not classified

Components:

2-Methylisothiazolin-3-one:

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

2-Methylisothiazolin-3-one:

Species	:	Rat
NOAEL	:	25 mg/kg
Application Route	:	Oral
Exposure time	:	90 d
Remarks	:	By analogy with a product of similar composition
Repeated dose toxicity - Assessment	:	Toxic if swallowed., Fatal if inhaled., Causes severe skin burns and eye damage.

Aspiration toxicity

Not classified

Components:

2-Methylisothiazolin-3-one:

No aspiration toxicity classification

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to fish	:	LC50 : 10 - 100 mg/l Method: calculated

Components:

5-Chloro-2-methyl-2H-isothiazol-3-one:

Ecotoxicology Assessment

Acute aquatic toxicity	:	Very toxic to aquatic life.
Chronic aquatic toxicity	:	Very toxic to aquatic life with long lasting effects.

2-Methylisothiazolin-3-one:

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): 4,77 mg/l Exposure time: 96 h

		Method: OECD Test Guideline 203
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 0,934 mg/l End point: mortality Exposure time: 48 h Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	:	NOEC (Pseudokirchneriella subcapitata (green algae)): 0,0104 mg/l End point: Biomass Exposure time: 96 h Method: OECD Test Guideline 201 EC50 (Pseudokirchneriella subcapitata (algae)): 0,063 mg/l End point: Biomass Exposure time: 96 h Method: OECD Test Guideline 201
M-Factor (Acute aquatic toxicity)	:	10
Toxicity to fish (Chronic toxicity)	:	Remarks: no data available
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	Remarks: no data available
M-Factor (Chronic aquatic toxicity)	:	1
Toxicity to microorganisms	:	EC50 (Bacteria): 31,7 mg/l Exposure time: 3 h

Ecotoxicology Assessment

Acute aquatic toxicity	:	Very toxic to aquatic life.
Chronic aquatic toxicity	:	Very toxic to aquatic life with long lasting effects.

Copper sulphate:

Ecotoxicology Assessment

Acute aquatic toxicity	:	Very toxic to aquatic life.
Chronic aquatic toxicity	:	Very toxic to aquatic life with long lasting effects.

Persistence and degradability

Components:

2-Methylisothiazolin-3-one:

Biodegradability	:	aerobic Result: Not rapidly biodegradable

Bioaccumulative potential

<u>Components:</u>

2-Methylisothiazolin-3-one:

Bioaccumulation	:	Remarks: Due to the distribution coefficient n-octanol/water, accumulation in organisms is not expected.

Mobility in soil

<u>Components:</u>

2-Methylisothiazolin-3-one:

Distribution among environmental compartments	:	Remarks: no data available

Other adverse effects

<u>Product:</u>

Additional ecological information	:	The product should not be allowed to enter drains, water courses or the soil. The classification was made by the conventional (calculation) method of the CLP Regulation (EC) No 1272/2008.

<u>Components:</u>

2-Methylisothiazolin-3-one:

Environmental fate and pathways	:	no data available

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	:	In accordance with regulations for hazardous waste, must be taken to a hazardous waste disposal site Product should be taken to a suitable and authorized waste disposal site in accordance with relevant regulations and if necessary after consultation with the waste disposal operator and/or the competent Authorities
Contaminated packaging	:	Regulations concerning reuse or disposal of used packaging materials must be observed. Packaging that cannot be cleaned should be disposed of as product waste

14. TRANSPORT INFORMATION

IATA

Proper shipping name:	Corrosive liquid, acidic, organic, n.o.s.
Class:	8
Packing group:	III
UN/ID number:	UN 3265
Primary risk:	8
Remarks	Shipment permitted
Hazard inducer(s):	Isothiazolinones

IMDG

Proper shipping name:	Corrosive liquid, acidic, organic, n.o.s.
Class:	8
Packing group:	III
UN no.	UN 3265
Primary risk:	8
Remarks	Shipment permitted
Hazard inducer(s):	Isothiazolinones
EmS :	F-A S-B

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Apart from the data/regulations specified in this chapter, no further information is available concerning safety, health and environmental protection.

Minister of Industry Regulation No. 23/M-IND/PER/4/2013 concerning the Revision of Minister of Industry Regulation No. 87/M-IND/PER/9/2009 concerning Globally Harmonized System of Classification and Labelling of Chemicals.

Regulation of the Minister of Health No. 472 of 1996 on the Safeguarding of Substances Hazardous to Health

Hazardous substances that must be registered : Not applicable

Government Regulation No. 74 of 2001 on the Management of Hazardous and Toxic Substances

Hazardous substances approved for use : Not applicable

Prohibited substances : Not applicable

Restricted substances : Not applicable

16. OTHER INFORMATION

Revision Date : 18.03.2026

Further information

Other information : Observe national and local legal requirements

Date format : yyyy/mm/dd

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MERCOSUR - The Agreement for the Facilitation of the Transport of Dangerous Goods; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

ID / EN